

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

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ASSESSMENT TOOL 1 – Data Interpretation

Course Code: DSA0501

Course Title: Query Processing for Data Science For Data Science

CO Assessed: CO1

Assessment: Tool 1 – Data Interpretation

Weightage:

Total Marks: 20 Marks

CO1 Statement: Ability to read, interpret, and process structured data using Python
Assessment: AT1 – Data Interpretation
Duration: [Insert duration, e.g., 1 hour]
Maximum Marks: 20

Student Name: P. Surayana

Reg. No.: 191414303

Section / Batch: AIGDS
Slot A (1014)

Date of Submission: 20/07/2020

Faculty In-charge: K. Veena Devi

Project Mode: Individual Submission

Code of Conduct

I, P. Surayana (191414303) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: P. Surayana

Data Loading & Setup (5Marks)	Correct Logic/Approach (5 Marks)	Accuracy of Final Output (5 Marks)	Output Presentation & Explanation (5Marks)	Total (20 Marks)
5	5	5	4	19

1. Employee Salary Analysis

EmployeeID	Name	Department	Salary
101	Arun Kumar	IT	55000
102	Priya Sharma	HR	42000
103	Rahul Verma	IT	61000
104	Sneha Iyer	Finance	48000
105	Vikram Singh	HR	53000
106	Anita Desai	Finance	72000
107	Karthik Raj	IT	39000
108	Divya Menon	Finance	58000
109	Suresh Nair	HR	45000
110	Meera Pillai	IT	67000

Employee Salary Analysis — display employees earning more than ₹50,000 and calculate department-wise average salary

2. Duplicate Record Removal

CustomerID	Name	Email	Phone
1	Rohit Sharma	rohit.sharma@email.com	9876543210
2	Anjali Gupta	anjali.gupta@email.com	9876543211
3	Manoj Tiwari	manoj.tiwari@email.com	9876543212
1	Rohit Sharma	rohit.sharma@email.com	9876543210
4	Kavita Rao	kavita.rao@email.com	9876543213
2	Anjali Gupta	anjali.gupta@email.com	9876543211
5	Deepak Joshi	deepak.joshi@email.com	9876543214
6	Sunita Reddy	sunita.reddy@email.com	9876543215
3	Manoj Tiwari	manoj.tiwari@email.com	9876543212
7	Farhan Ali	farhan.ali@email.com	9876543216

Duplicate Record Removal — find and remove duplicate customer records, display the cleaned dataset

3. Sorting Dataset

ProductID	ProductName	SalesAmount
P01	Wireless Mouse	45000
P02	Bluetooth Speaker	72000
P03	Laptop Stand	28000
P04	Mechanical Keyboard	91000
P05	USB-C Hub	33000
P06	Webcam HD	58000

P07	Gaming Headset	84000
P08	Portable SSD	105000
P09	Smartwatch	120000
P10	Power Bank	39000
P11	Desk Lamp	21000
P12	Monitor Stand	26000
P13	Noise Cancelling Earbuds	97000
P14	Router	47000
P15	Graphic Tablet	63000

Sorting Dataset — sort products by sales amount and display the top 10 highest-selling products

4. Read JSON Data

```
[
  {
    "StudentID": 1,
    "Name": "Aditi Rao",
    "Marks": 88
  },
  {
    "StudentID": 2,
    "Name": "Bhavesh Patel",
    "Marks": 92
  },
  {
    "StudentID": 3,
    "Name": "Chitra Nair",
    "Marks": 79
  },
  {
    "StudentID": 4,
    "Name": "Dev Malhotra",
    "Marks": 95
  },
  {
    "StudentID": 5,
    "Name": "Esha Kapoor",
    "Marks": 84
  },
  {
    "StudentID": 6,
    "Name": "Faisal Khan",
    "Marks": 90
  },
  {
    "StudentID": 7,
    "Name": "Gauri Deshmukh",
    "Marks": 73
  },
  {
    "StudentID": 8,
    "Name": "Harish Chandran",
    "Marks": 88
  }
]
```

Read JSON Data — read student data, display all student names, and identify the topper

5.

Month	SalesAmount
January	52000
February	61000
March	47000
April	73000
May	68000
June	55000
July	80000
August	62000
September	58000
October	91000
November	76000
December	99000

Monthly Sales Aggregation, Given monthly sales data

Task

- Read the dataset.
- Calculate total sales.
- Calculate average sales.
- Calculate maximum sales.
- Calculate minimum sales.

P. Surayana.

191424303.

Employee Salary Analysis :-

Aim:- Read employee data, display employees earning more than ₹50,000 and calculate department-wise average salary.

Python Code:- Import pandas as pd

```
employee = {
```

```
    "Employee ID" = [101, 102, 103, 104, 105, 106, 107, 108]
```

```
    "Name" : ["Arun", "Kumar", "Priya", "Rahul", "Verna",  
             "Sneha", "Iyer", "Mary"],
```

```
    "Dept" : ["IT", "HR", "IT", "HR", "HR", "IT", "HR", "IT"]
```

```
    "Salary" : [55000, 42000, 61000, 60000, 42000,  
                80,000, 90,000, 62000], }
```

```
df = pd.DataFrame(employee)
```

```
print("Employee Dataset")
```

```
print(df)
```

```
print("\n Employees earning more than ₹50,000")
```

```
print(df[df["salary"] > 50000])
```

```
print("\n Department-wise Average Salary")
```

```
print(df.groupby("Department")["salary"].mean())
```

Output:-

Employees earning above ₹50,000 :-

Arun, Priya, Rahul, Verna, Sneha, Iyer, Mary.

Department-wise Average Salary:-

Finance

HR

IT

0

466666.67

55500.00

Explanation:-

- 1) Employee data is loaded into a dataframe.
- 2) Employees with salary greater than ₹50,000 are filtered.
- 3) Average salary is calculated using `groupby()`.

2) Duplicate Record Removal:-

Aim:- Remove duplicate customer records.

Python:- `import pandas as pd`

```
customer = {
```

```
    "customer ID": [1, 2, 3, 1, 4, 2, 5, 6],
```

```
    "Name": ["Rohit Sharma", "Anjali Gupta", "Manoj",  
            "Rohit Sharma", "Kavitha", "Anjali Gupta",
```

```
            "Deepak", "Sunita"],
```

```
    "Email": ["rohit.sharma@gmail.com",
```

```
            "Anjali.gupta@gmail.com",
```

```
            "Manoj@gmail.com",
```

```
            "rohit.sharma@gmail.com",
```

```
            "Kavitha@gmail.com",
```

```
            "Anjali.gupta@gmail.com",
```

```
            "deepak@gmail.com",
```

```
            "Sunita@gmail.com"]}]
```

```
df = pd.DataFrame(customer)
```

```
print("Original Dataset")
```

```
print(df)
```

```
clean = df.drop_duplicates()
```

```
print("\n Dataset after removing Duplicates")  
print(clean)
```


Output:-

Total Records Before = 10,

Total Records after = 7.

Explanation:-

1) Dataset is loaded.

2) drop_duplicates() removes repeated customer records.

3) clean dataset is displayed.

3) Sorting Dataset:-

Aim:- Sort products based on sales amount and display the top 10.

Python code:- import pandas as pd

```
product = {  
    "productID": ["P01", "P02", "P03", "P04", "P05", "P06", "P07",  
                  "P08"],  
    "ProductName": ["Wireless Mouse", "Bluetooth Speaker",  
                    "Laptop Stand", "Mechanical keyboard",  
                    "Desk lamp", "Monitor stand", "Router",  
                    "PowerBank"],  
    "SalesAmount": [45000, 72000, 28000, 91000, 33000,  
                    58000, 120000, 63000],  
}
```

```
df = pd.DataFrame(product)
```

```
top 8 = df.sort_values(by="SalesAmount", ascending =  
                        False).head(10)
```

```
Print (top 10).
```

Output:-

Rank	Product	Sales
1.	Wireless mouse	45000
2.	Bluetooth speaker	42000
3.	laptop stand	81000
4.	Mechanical keyboard	91000
5.	Desk lamp	45000
6.	Monitor stand	68000
7.	Router	47000
8.	PowerBank	63000

Explanation:-

- 1) Dataset is sorted in descending order.
- 2) head(10) displays the highest-selling ten products.

4) Read JSON Data :-

Aim:- Read JSON data, display all student names and identify the topper.

python code:- import pandas as pd

```
students = [
    {"student_id": 1, "Name": "Aditi", "Marks": 80},
    {"student_id": 2, "Name": "Bhavesh", "Marks": 92},
    {"student_id": 3, "Name": "Akash", "Marks": 93},
    {"student_id": 4, "Name": "Harish", "Marks": 84},
]
```

```
df = pd.DataFrame(students)
```

```
print(df["Name"])
```



```
topper = df.loc[df["Marks"].idxmax()]
```

```
print("\n Topper ")
```

```
print ( " topper ")
```

Output :- Student Names :-

1) Aditi.

2) Bhavesh.

3) Akash.

4) Harish.

Topper :- Akash

Marks = 93.

Explanation:-

1) JSON data is converted into a DataFrame.

2) Student Names are displayed.

3) idxmax() identifies the student with the highest marks.

5) Monthly Sales Aggregation :-

Aim:- Calculate total, average, maximum and minimum sales.

python code:- import pandas as pd.

```
sales = {  
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul",  
              "Aug", "Sep", "Oct", "Nov", "Dec"],
```

```
    "Sales_Amount": [51000, 64000, 47000, 73000, 68000,  
                     55000, 80000, 61000, 58000, 91000, 78000, 99000]
```

```
}
```

```
df = pd.DataFrame(sales)
```

```
print ("total sales = ", df["Sales_Amount"].sum())
```

```
print ("Average sales = ", df["Sales_Amount"].mean())
```

```
print ("Maximum sales=", dt["Sales Amount"].max())
print ("Minimum sales=", dt["Sales Amount"].min())
```

Output:-

Calculation

Total Sales

Average sales

Maximum sales

Minimum sales

Result

Total Sales: ₹ 822000

Average sales: ₹ 685000

Maximum sales: ₹ 99000

Minimum sales: ₹ 47000

Explanation:-

- 1) The sales dataset is loaded into DataFrame.
- 2) Pandas aggregation functions (sum(), mean(), max(), min()) are used to compute the required statistics.
- 3) The calculated values are displayed clearly.